



Independent Study Proposal

Emerging Media System Design for Responsive Theatre: TouchDesigner,
QLab, and the Self-Aware Performance Environment

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Department: Entertainment Technology

Sponsoring Organization: Center for Holistic Integration (CHI)

Project Context: Responsive Theatre Development using Touch Designer

Project Overview

This independent study will investigate the technical and theoretical infrastructure required to support a future self-referential culmination presentation on Emerging Media Technology in theatre. The proposed culmination project is envisioned as a TED-style lecture-performance that demonstrates, through its own self-referential form, how emerging media systems can be embedded within a traditional theatrical environment. However, this independent study is not primarily responsible for producing the final lecture, script, or performance content. Instead, Gabriel Aguilar's work will focus on researching the systems that would make such a presentation technically and conceptually possible.

The study explores how TouchDesigner might function as a central mediation environment within a responsive theatrical control ecology. TouchDesigner will be examined not as a replacement for established theatrical systems such as QLab, lighting consoles, sound playback systems, projection systems, or show-control environments, but as a possible integration layer capable of receiving, processing, transforming, and redistributing live data. The project will investigate how QLab, TouchDesigner, sensors, AI-supported semantic evaluation, and human operators might communicate within a unified theatrical workflow.

The broader research question is how Emerging Media Design can be understood as a distinct theatrical design discipline. Traditional theatre already recognizes scenery, costume, lighting, sound, video, and show control as design or production domains with relatively established responsibilities and workflows. Emerging Media work, by contrast, is often treated as a special effect, a technical add-on, or the responsibility of whoever happens to know a particular software environment. This study will test the premise that responsive, interactive, immersive, AI-mediated, and semantically aware theatrical

systems require their own design logic, documentation practices, authority structures, and technical workflows.

Central Concept

The central concept of this independent study is that Gabriel will research and prototype the workflow infrastructure for a self-aware theatrical presentation environment. The project may or may not produce finished media content for the later culmination presentation. Its primary deliverable is a technical and theoretical design package that clarifies what such a system would require, what TouchDesigner can and cannot provide, how it might integrate with QLab and other systems, and what additional tools, protocols, or design practices would be needed.

In this context, “self-aware” does not imply consciousness. It refers to a performance environment capable of sensing selected aspects of its own state, representing those states in a usable form, and making bounded decisions or recommendations based on those representations. A self-aware room might know, for example, whether it is in lecture mode, demonstration mode, audience-response mode, manual override, or system fault. It might receive data from microphones, cameras, sensors, cue states, performer actions, or operator input. It might then transform those inputs into media changes, cue triggers, visualizations, logs, or prompts for human decision-making.

The study will therefore treat the theatre not as a passive container for technology, but as an active blended environment in which physical space, virtual systems, conceptual meanings, human performers, computational agents, and traditional production departments interact.

Relationship to CHI, BBS, and BRPS

This independent study will be conducted under the aegis of the Center for Holistic Integration (CHI). CHI supports modular, scalable, interdisciplinary projects that can contribute to larger meta-projects without requiring every participant to produce the final form of the larger project. Gabriel’s independent study should be understood as one such modular contribution: it develops technical and conceptual infrastructure that may later support a culmination project, a Blended Reality Performance System prototype, or a broader CHI smart-room initiative.

The study also connects to the Balanced Blended Space (BBS) framework. BBS provides a way to analyze mediation among physical, virtual, and conceptual spaces, and among cognitive and computational agents. In this project, BBS can help describe how a lecturer, audience, room, sensors, TouchDesigner network, QLab cue stack, AI evaluator, and human operator participate in a mediated performance system.

The project also supports the broader Blended Reality Performance System (BRPS) research direction. BRPS requires practical workflows for integrating physical performance, virtual media, real-time control, and computational decision-making. Gabriel's study can contribute to that effort by testing specific signal pathways and by documenting the design assumptions, limitations, and failure points that emerge.

Primary Research Question

How can TouchDesigner function as a mediation layer within a responsive theatrical environment, and what additional workflows, protocols, semantic models, and governance structures are needed to support Emerging Media Design as a distinct theatrical discipline?

Supporting Questions

- What theatrical functions are best handled by QLab, TouchDesigner, lighting consoles, sound systems, projection systems, or other specialized environments?
- How can OSC, MIDI, network protocols, or other communication methods allow these systems to exchange information reliably?
- What kinds of room-state information can be sensed, inferred, represented, or logged?
- What does TouchDesigner do well in a live theatrical context, and where does it require support from other tools or workflows?
- How can AI or rule-based logic contribute to semantic evaluation of room state without becoming an uncontrolled show operator?
- What documentation would allow another student, designer, or technician to understand and continue the system?
- How should authority, control, observability, distortion, and effort be distributed across the human and computational agents in the system?

Scope of Work

This independent study will not attempt to build a complete production system. It will instead develop a set of focused research and prototype components that test the viability of a future self-aware performance environment.

Gabriel will investigate TouchDesigner's use as a real-time media and interaction platform, with particular attention to its possible integration with QLab. The study may include OSC communication, audio analysis, camera or sensor input, projection or visual response, state-machine logic, and basic documentation of communication pathways. If feasible, Gabriel may also test a simple AI-supported or rule-based semantic evaluation layer, but the project should not depend on complex AI integration in order to be successful.

The project should remain architectural rather than content-driven. Visual or sonic examples may be created for testing, but they do not need to become final aesthetic material for the culmination presentation. The goal is to clarify throughput: how information, cues, signals, decisions, authority, and documentation move through the system.

Proposed Technical Architecture

A tentative architecture for the study may include the following components:

- QLab as the conventional theatrical cue playback and show-control environment.
- TouchDesigner as the real-time visual, sensor, interaction, and mediation environment.
- OSC as the primary communication protocol between QLab and TouchDesigner.
- Microphones, camera input, keyboard input, MIDI controllers, or other available sensors as test inputs.
- A rule-based or lightweight semantic-state model that classifies the room into states such as idle, lecture, demonstration, audience response, manual override, and fault.
- Optional AI-assisted evaluation, if technically feasible and appropriate within the available time.
- A human operator or stage manager layer with explicit authority to trigger, confirm, or override system behavior.
- A documentation and logging layer that records tests, system decisions, known issues, and development notes.

This architecture should remain flexible. One purpose of the study is to determine whether this structure is practical, whether TouchDesigner should remain central, and where additional tools or middleware may be required.

Proposed Room-State Vocabulary

Gabriel will develop and test a preliminary vocabulary for describing the state of a responsive performance environment. The following states are suggested as a starting point:

Idle State: The system is active but not currently responding to performance input.

Lecture State: The room prioritizes clarity, stability, and controlled media support.

Demonstration State: The room allows visible interaction between speaker, media, and system response.

Audience Response State: Audience sound, movement, or interaction may influence selected media parameters.

Semantic Evaluation State: The system attempts to classify a higher-level condition, such as engagement, transition, uncertainty, or cue readiness.

Manual Override State: Human authority suspends or redirects automated system behavior.

Fault or Uncertainty State: The system cannot confidently interpret input or execute an action safely, and should fail into a stable condition.

This vocabulary is not final. A major task of the study will be to determine whether these states are useful, too vague, too complex, or technically impractical.

Theoretical Frameworks

The study will use several conceptual frameworks to evaluate the system.

First, the project will use CHI's meta-project model. Gabriel's work should be treated as a modular contribution to a larger research ecosystem rather than as a stand-alone production assignment.

Second, the project will use BBS to map mediation pathways among physical, virtual, and conceptual spaces. For example, a sound in the physical room may be captured by a microphone, converted into virtual data, interpreted conceptually as audience response, and then returned to the room as a projected visual or cue change.

Third, the project will use EDOCA as an evaluative tool. Gabriel should consider Effort, Distortion, Observability, Control, and Authority throughout the system. This is especially important if AI or semantic evaluation is introduced. The project should ask who can observe system behavior, who can control it, what distortion occurs during interpretation, how much effort is required to operate or troubleshoot it, and who has authority to approve or override computational decisions.

Practical Deliverables

The final practical deliverables should include:

- A working TouchDesigner prototype or set of prototype networks demonstrating selected interaction pathways.
- A QLab-to-TouchDesigner communication test, preferably using OSC.
- A system architecture diagram showing major components and signal flow.
- A preliminary room-state model and state-transition diagram.

- A control authority matrix describing which human or computational agent can trigger, approve, modify, override, or log specific actions.
- A technical workflow document explaining setup, operation, dependencies, naming conventions, and known issues.
- A limitations report identifying what TouchDesigner handles well, what it does not handle well, and what additional systems or protocols may be required.
- A short demonstration showing at least one successful communication pathway and one example of state-based media response.

Theoretical Deliverables

The final theoretical deliverables should include:

- A short position paper defining Emerging Media Design as a theatrical design discipline.
- A BBS-style mediation analysis of the prototype system.
- An EDOCA analysis of at least one tested interaction pathway.
- A reflection on the role of TouchDesigner as a mediation environment rather than a complete theatrical control solution.
- A discussion of the potential and limits of AI as a semantic evaluator within a live performance environment.
- A future-development roadmap identifying how this work could support the later culmination project.

Suggested Semester Timeline

4.1 Summer Session

During the summer session, Gabriel will establish the project's technical and research infrastructure, including documentation repositories, software environments, and relevant workflow resources. He will investigate TouchDesigner, QLab, OSC communication, sensor and input possibilities, and related examples of interactive and responsive theatrical systems. At the same time, he will begin mapping the project's conceptual and technical architecture, testing prototype communication pathways, and developing preliminary documentation practices for the study.

The primary objective of the summer session is to establish a strong foundation for the project while generating meaningful technical, creative, and scholarly outcomes. Significant progress is expected in research, systems analysis, prototype development, workflow testing, and project documentation. Depending upon project development and available time, some practical and theoretical deliverables may reach completion during this period.

4.2 Extension Through Incomplete (If Required)

Should additional time be required to achieve the intended technical, scholarly, or documentation goals of the project, work may continue through an approved Incomplete in accordance with college policy. During this period, Gabriel may continue prototype development, expand research and testing activities, refine documentation, complete diagrams and written analyses, and finalize project deliverables.

Documentation Expectations

All work should be documented continuously. Gabriel should maintain a project log that records weekly goals, tests performed, problems encountered, solutions attempted, and next steps. Documentation should include screenshots, diagrams, code or patch notes, version information, and clear explanations of how each prototype works.

If appropriate, documentation should be organized in a GitHub repository or other CHI-approved documentation space. The repository should separate research notes, technical diagrams, TouchDesigner files, QLab files, media assets, and final reports. File naming conventions should be clear enough that future students can continue the work.

Assessment Criteria

The independent study should be evaluated according to the following criteria:

Conceptual Clarity: Gabriel clearly explains the role of Emerging Media Design and distinguishes it from conventional theatre departments and generic software operation.

System Architecture: Gabriel produces a coherent, modular, and well-documented architecture for a responsive theatrical environment.

Prototype Functionality: Gabriel demonstrates at least one working communication pathway between systems and at least one example of state-based response.

Documentation Quality: Gabriel produces documentation that another student or production team could reasonably understand and continue.

Critical Assessment: Gabriel identifies limitations, fragilities, missing capabilities, and future development needs honestly and specifically.

CHI/BBS Alignment: Gabriel analyzes the system as a mediation environment involving physical, virtual, and conceptual spaces as well as human and computational agents.

EDOCA Awareness: Gabriel addresses effort, distortion, observability, control, and authority in the design and assessment of the prototype.

Boundaries and Non-Goals

This independent study is not responsible for producing the final TED-style culmination presentation. It is not required to generate polished media content, a final script, a finished projection design, a complete performance system, or a production-ready installation. Those may emerge later from this work, but they are not the primary goals of the study.

The study is also not intended to prove that TouchDesigner is the only or best solution. A successful outcome may identify TouchDesigner's limitations and recommend additional tools, middleware, or alternate workflows. Negative findings are valid research outcomes if they are carefully documented and assessed.

Final Outcome

The expected final outcome is a reusable technical and theoretical design package for a future self-aware lecture-performance environment. This package should help future students, designers, and faculty understand how QLab, TouchDesigner, sensors, AI-supported semantic evaluation, and human operators might be integrated into a responsive theatrical workflow.

The strongest version of this independent study will leave behind not only a prototype, but also a clearer production language for Emerging Media Design as a distinct theatrical discipline.

Faculty Evaluation of Project Value

This project provides the student with an opportunity to integrate technical research, systems design, documentation, and emerging media practice within a structured independent-study environment. Through the investigation of TouchDesigner, QLab, OSC communication, responsive theatrical workflows, and the conceptual foundations of Emerging Media Design, the project supports both technical development and scholarly engagement.

The study aligns with the educational objectives of the Emerging Media Technology program while contributing to ongoing research and development activities within CHI, particularly in relation to responsive performance systems, blended environments, and modular infrastructure for future theatrical experimentation. The resulting work is expected to strengthen Gabriel Aguilar's abilities in research, systems analysis, technical prototyping, documentation, critical evaluation, and professional project development while producing outcomes that may support future academic, artistic, and production-oriented work.